Methods for Variant Analysis I

Bio5488
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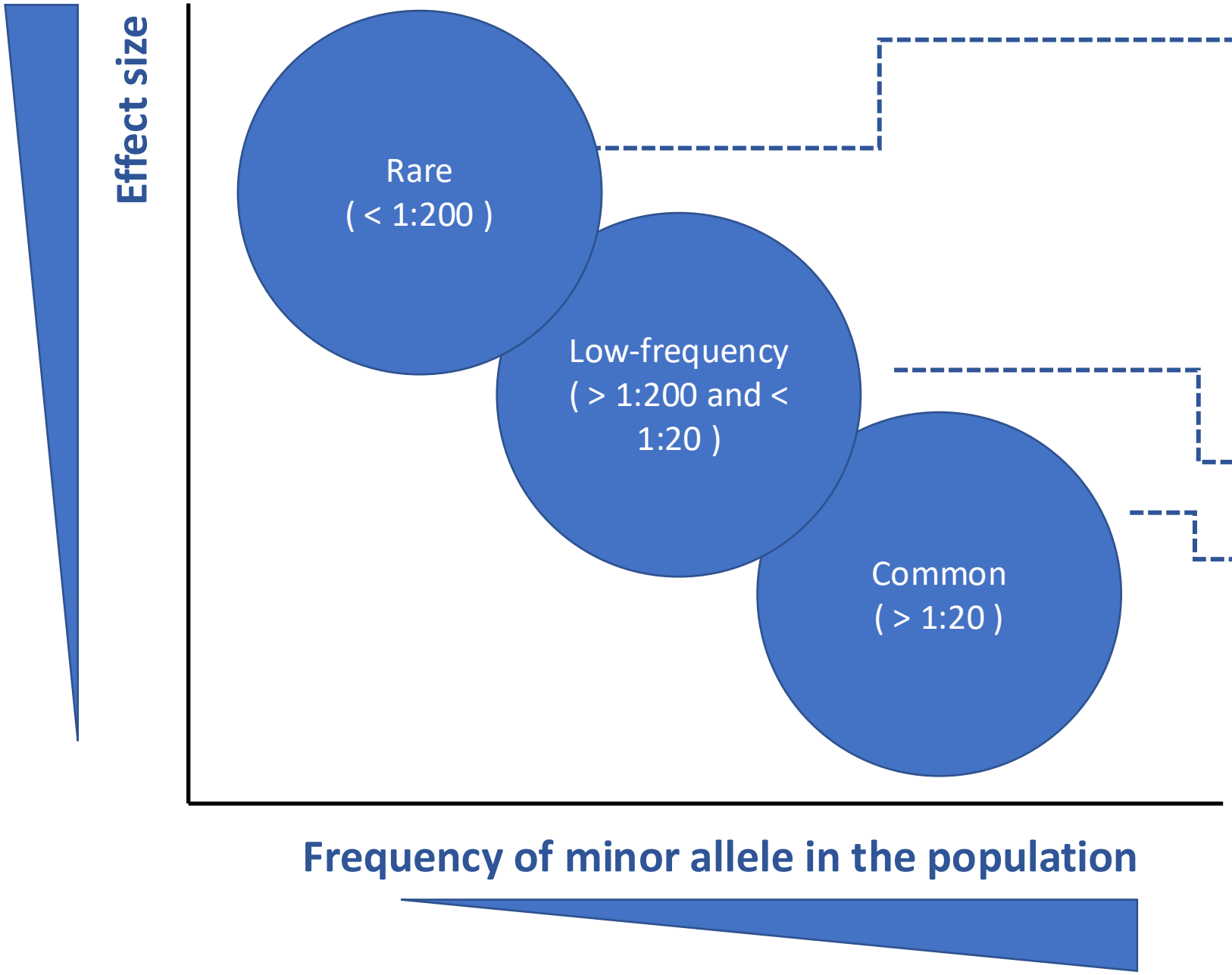
03/16/2026

WashU

Approaches to identify genetic variants by risk allele frequency



Long-read or short-read sequencing

Genotyping arrays:
- GWAS Chip
- Exome Chip

Modified from Manolio et al., *Nature* 2009

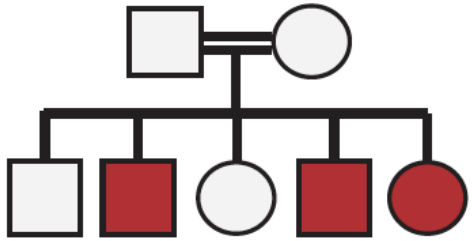
Study designs linking genotype with phenotype

Effect size

Sequencing as a discovery tool

March 18

Family-based studies

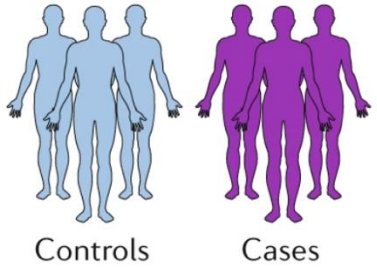


Extremes of a distribution

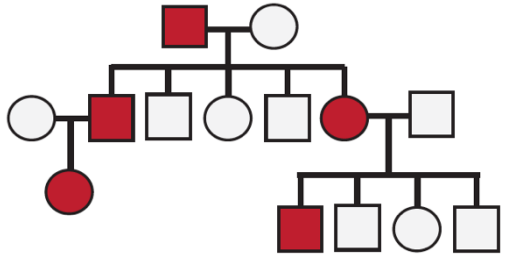
Genotyping as a discovery tool

March 16

Case-control association studies

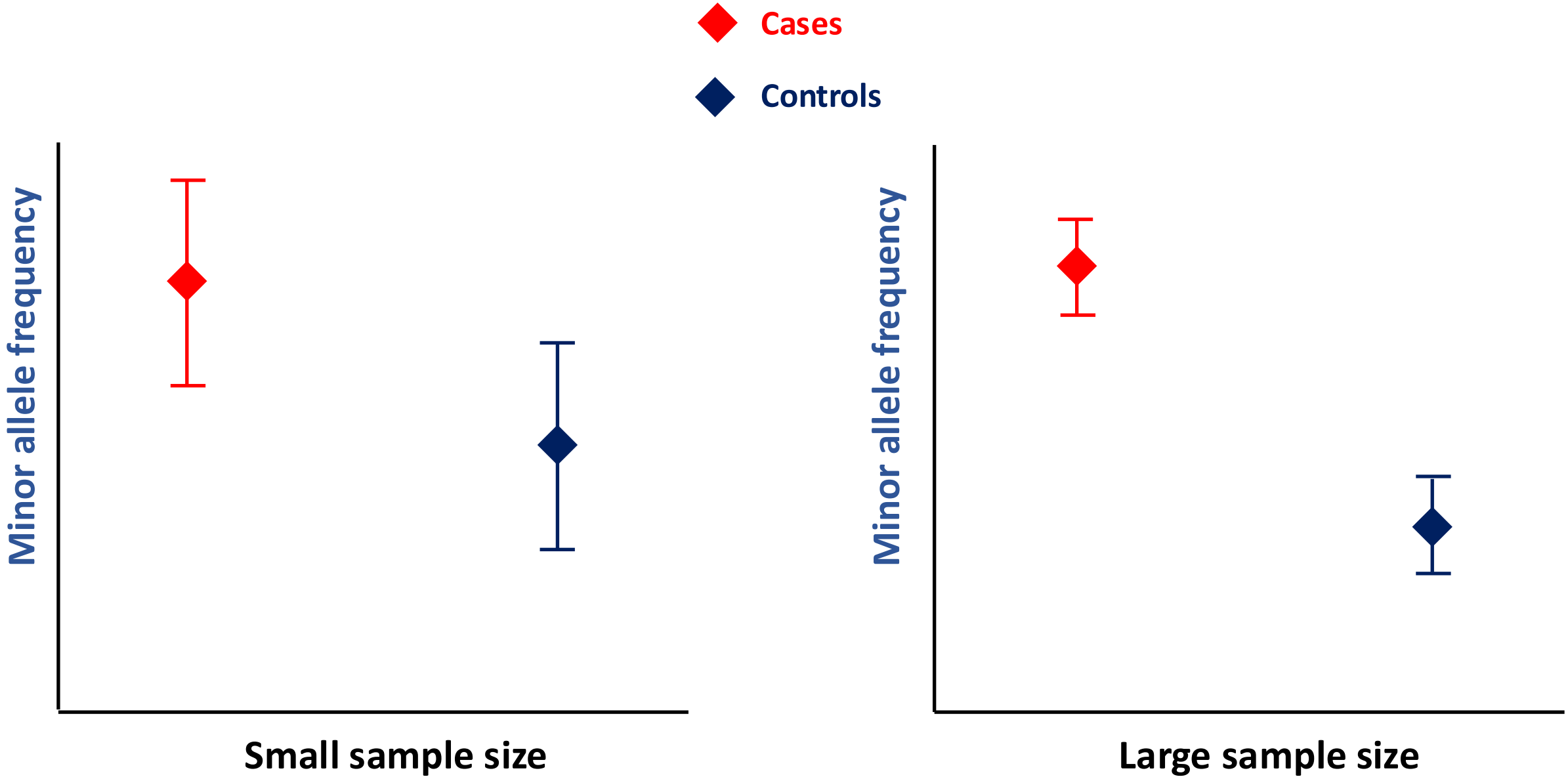


Linkage studies of multiplex families



Frequency of minor allele in the population

Statistical power for case-control association studies



Genome-wide association studies test the common disease – common variant hypothesis

- Single nucleotide polymorphism (SNPs)

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Haplotype 1: ATACGTATCGACA (75% frequency)
Haplotype 2: ACACGTGTCGACA (25% frequency)
```

- Hypothesis: Presence of SNP allele is associated with disease/trait
  - Is the C allele associated with trait?

Association tests in case-control samples

Most basic test for biallelic markers: compare allele frequencies in cases vs controls in a 2x2 table.

	# of minor alleles	# of major alleles	Total alleles
Cases	N_{11}	N_{12}	$2xN_{case}$
Controls	N_{21}	N_{22}	$2xN_{control}$

To detect association between genetic variant and disease risk, we can simply perform:

- Fisher's exact test or

- Chi-squared test

$$\chi^2 = \Sigma \frac{(obs - exp)^2}{exp}$$

Association tests in case-control samples (cont.)

Alternatives: logistic regression for binary traits

Y : 1 if case, 0 if control; P: probability that Y = 1 (case)

$$logit\ (P) = \ln\left(\frac{P}{1-P}\right) = \alpha + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Alternatives: linear regression for continuous traits

Y : quantitative trait value

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

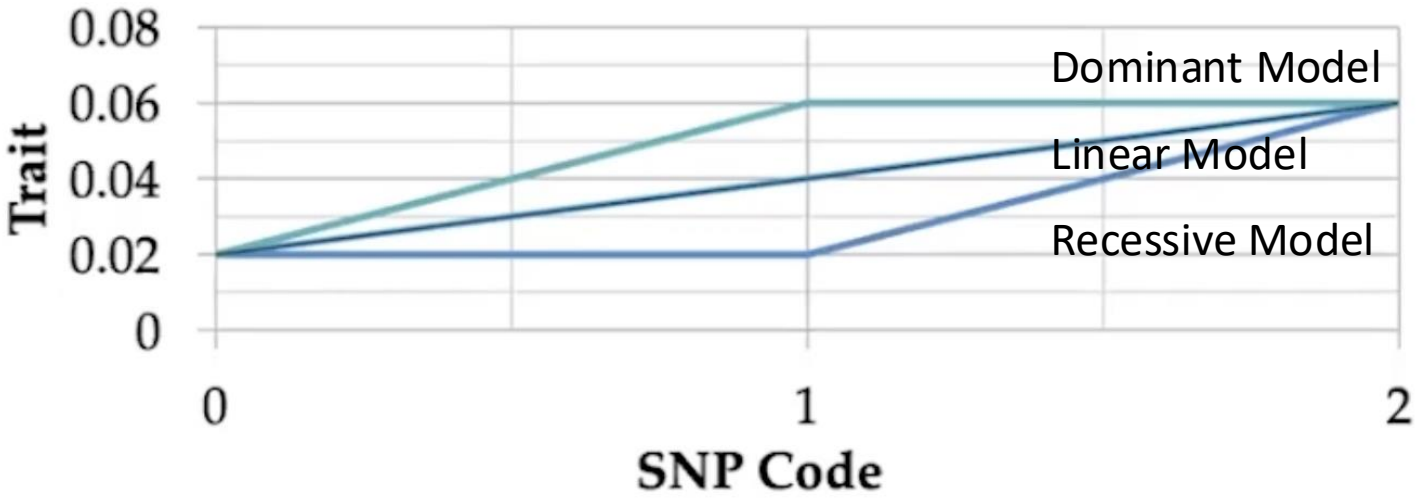
X_1: genotype (additive coding)
X_{2-n}: covariates (e.g., gender, age)

Modeling SNP effects in regression

Genetic Model	SNP code	Genotype
Additive	0	TT
	1	TC
	2	CC
Dominant	0	TT
	1	TC or CC
Recessive	0	TT or TC
	1	CC

```
Haplotype 1: ATACGTATCGACA (75% frequency)
Haplotype 2: ACACGTGTCGACA (25% frequency)
```

- Is the C allele associated with trait?



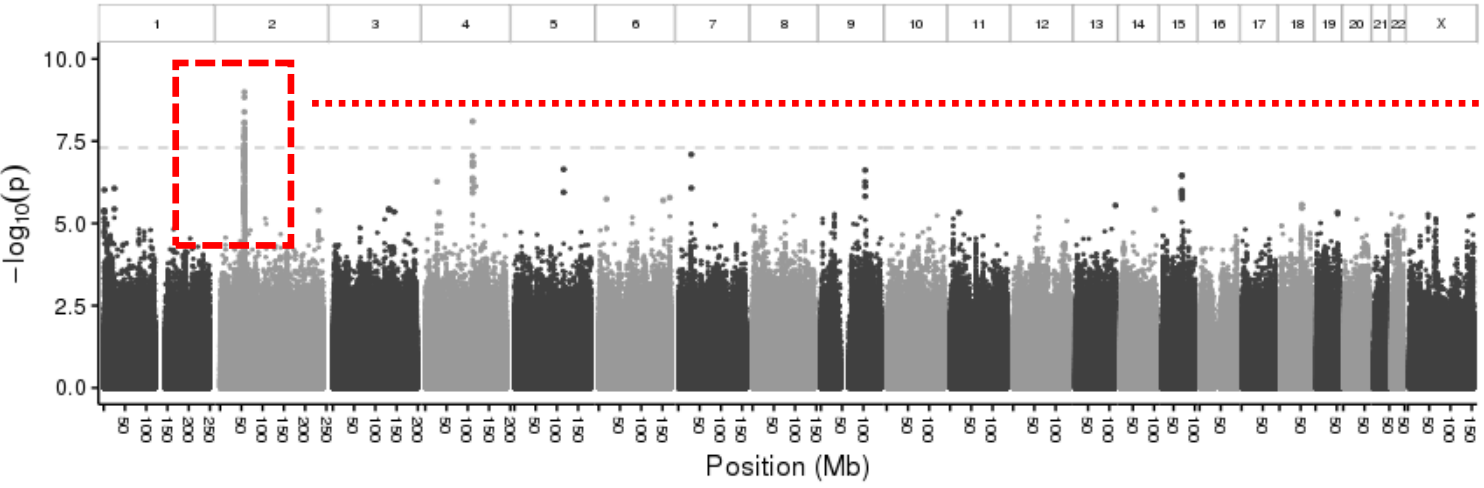
More nuanced statistical methods for case-control association study

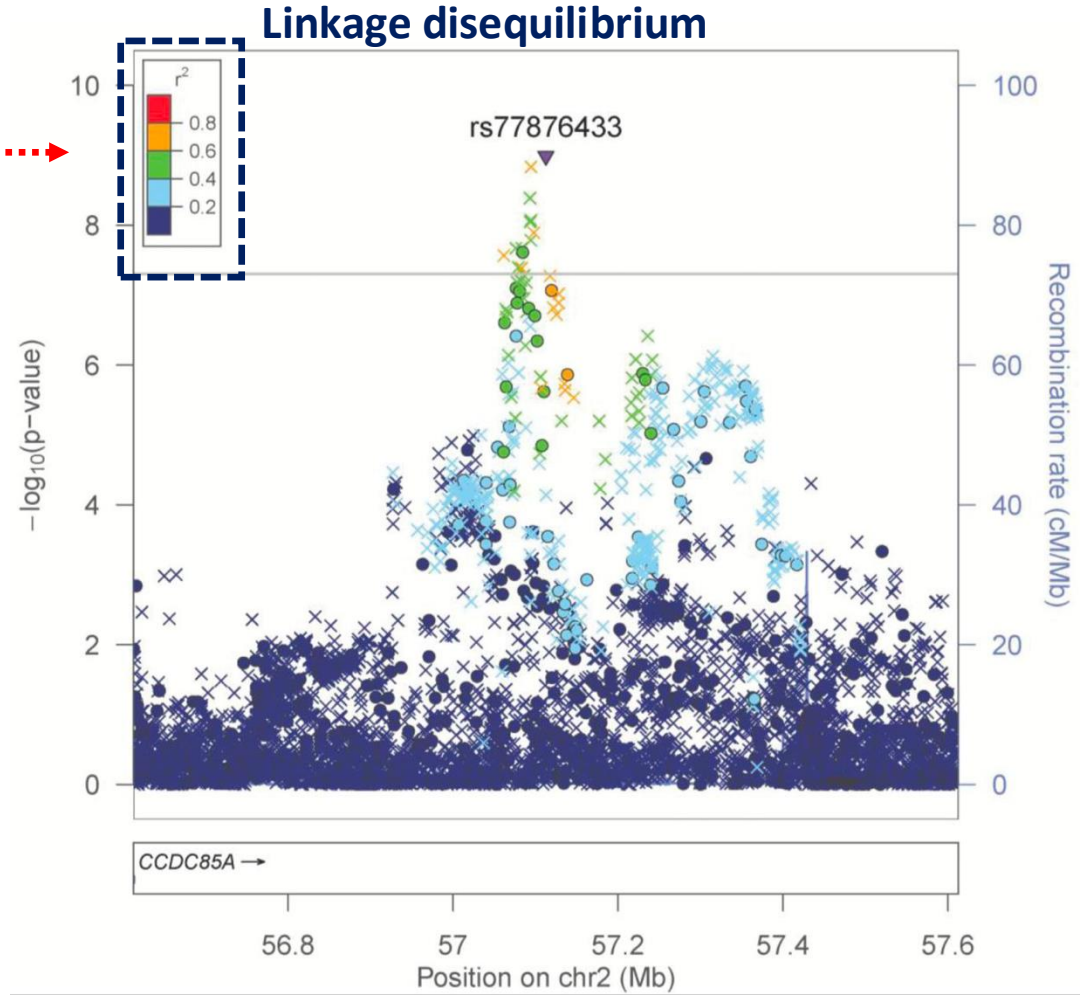
Table 2. Summary of Statistical Methods for Rare-Variant Association Testing

	Description	Methods	Advantage	Disadvantage	Software Packages[a]
Burden tests	collapse rare variants into genetic scores	ARIEL test,[50] CAST,[51] CMC method,[52] MZ test,[53] WSS[54]	are powerful when a large proportion of variants are causal and effects are in the same direction	lose power in the presence of both trait-increasing and trait-decreasing variants or a small fraction of causal variants	EPACTS, GRANVIL, PLINK/SEQ, Rvtests, SCORE-Seq, SKAT, VAT
Adaptive burden tests	use data-adaptive weights or thresholds	aSum,[55] Step-up,[56] EREC test,[57] VT,[58] KBAC method,[59] RBT[60]	are more robust than burden tests using fixed weights or thresholds; some tests can improve result interpretation	are often computationally intensive; VT requires the same assumptions as burden tests	EPACTS, KBAC, PLINK/SEQ, Rvtests, SCORE-Seq, VAT
Variance-component tests	test variance of genetic effects	SKAT,[61] SSU test,[62] C-alpha test[63]	are powerful in the presence of both trait-increasing and trait-decreasing variants or a small fraction of causal variants	are less powerful than burden tests when most variants are causal and effects are in the same direction	EPACTS, PLINK/SEQ, SCORE-Seq, SKAT, VAT
Combined tests	combine burden and variance-component tests	SKAT-O,[64] Fisher method,[65] MiST[66]	are more robust with respect to the percentage of causal variants and the presence of both trait-increasing and trait-decreasing variants	can be slightly less powerful than burden or variance-component tests if their assumptions are largely held; some methods (e.g., the Fisher method) are computationally intensive	EPACTS, PLINK/SEQ, MiST, SKAT
EC test	exponentially combines score statistics	EC test[67]	is powerful when a very small proportion of variants are causal	is computationally intensive; is less powerful when a moderate or large proportion of variants are causal	no software is available yet

Lee et al., *AJHG* 2014

Displaying association test results: Manhattan plot





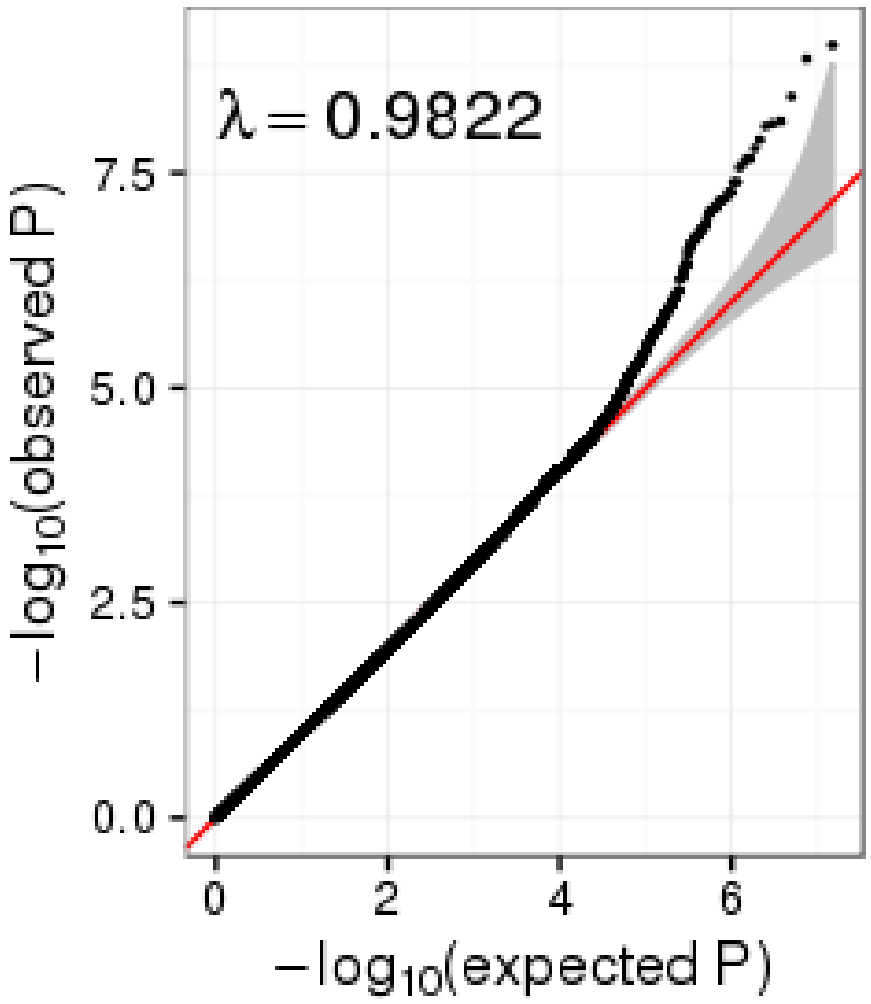
Non-daily vs daily smoking
778 cases, 1,786 controls

x-axis: chromosomal position
y-axis: $-\log_{10}$(p-value)

Circle: genotyped SNP
Cross: imputed SNP

Saccone et al., Nicotine and Tobacco Research 2016

Displaying association test results: Q-Q plot



Quantile-Quantile plot (Q-Q plot)

Idea: Rank tested SNPs by association evidence; compare the number of observed associations versus the expected associations under the null at a given significance level

Under the null, p-values follow a uniform distribution

λ: genomic inflation factor

Median of the observed chi-squared test statistics divided by the expected median of the corresponding chi-squared distribution; A λ value close to 1 indicates no evidence of inflation, while values up to 1.10 are generally considered appropriate for association studies

Saccone et al., Nicotine and Tobacco Research 2016

Displaying association test results: Q-Q plot (cont.)

Helps detect systematic bias in data:

- If no inflation, most data points should be close to the y=x line
- Exception: signals (lowest, most significant p-values)

Statistical tests: P-value

Definition: Probability of observing as strong of an effect if there is no association

- False positive rate α = 0.05
- If there is no association, 5% of the tests will have a p-value ≤ 0.05

Problem: When multiple tests are performed, the probability of having at least one false positive > α

Solution: New α to maintain an overall false positive rate of 0.05

- Number of independent tests being performed
- Bonferroni correction: p ≤ 0.05/# independent tests

In the GWAS setting –how to determine the number of "independent tests"?

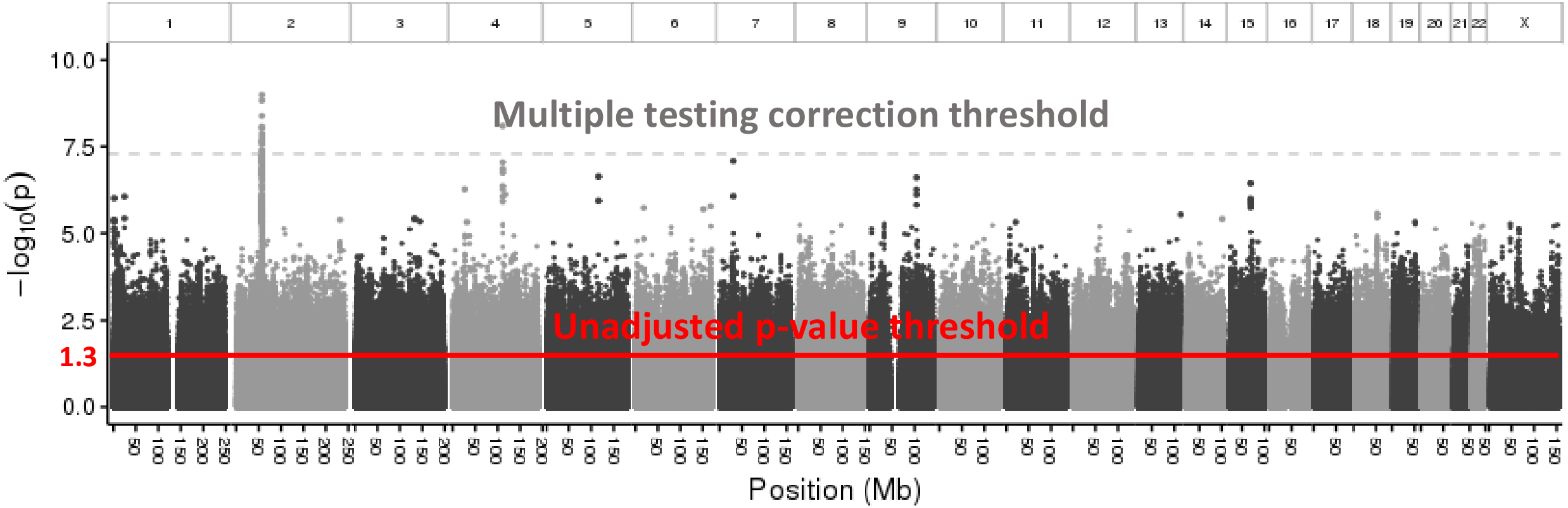
- Account for linkage disequilibrium, or correlation between alleles (more on this later)
- Typical threshold corrects for 1 million independent tests, requiring $p \leq 5 \times 10^{-8}$

Pe'er I et al. *Genet Epidemiol* 2008 Hoggart CJ et al. *Genet Epidemiol* 2008

Multiple testing correction

Why do we correct for multiple tests?

- Conventional GWAS significance threshold is $p \leq 5 \times 10^{-8}$
- Corresponding to $-\log(p) \sim 7.3$
- **In contrast: threshold of $p \leq 0.05$ corresponds to $-\log(p) \sim 1.3$**



Potential issues in association tests

Population stratification (Sul et al. *PLoS Genetics* 2018)

Sex-differential participation bias (Pirastu et al. *Nature Genetics* 2021)

Poorly matched cases and controls (e.g., age, case-control ratio, environmental exposure)

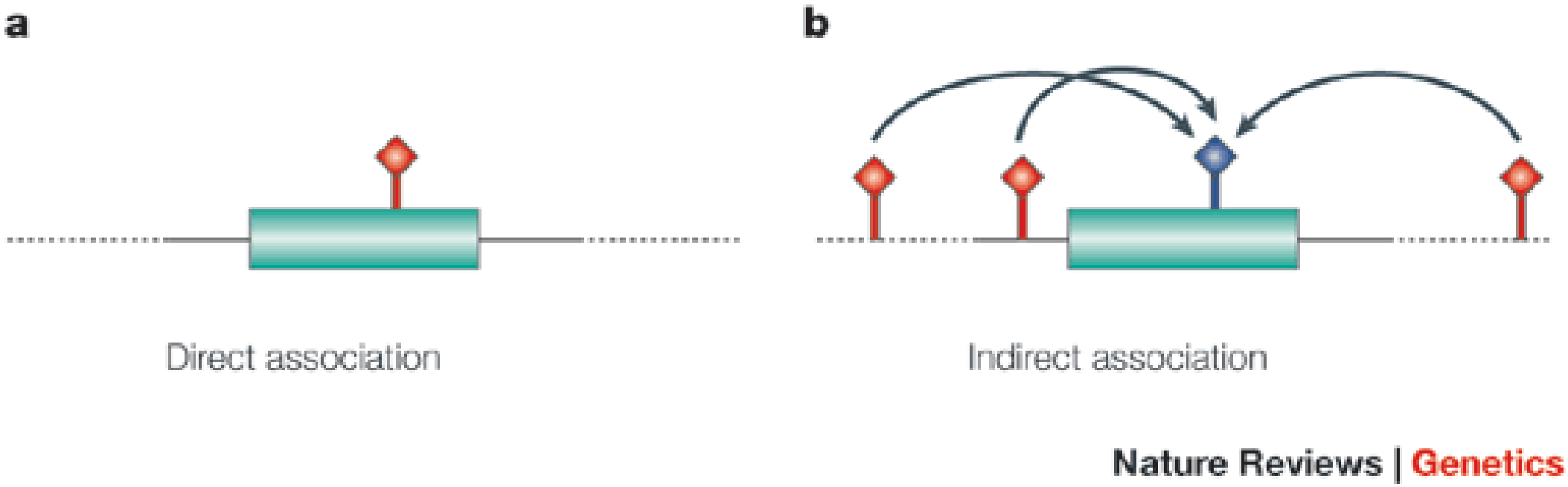
Not accounting for effects due to consanguinity and phenotypic heterogeneity (Jin et al. *Nature Genetics* 2017)

Inflation of false positive rates (multiple testing correction)

Hard to establish genome-wide significance when the total number of minor alleles is small (The p-value of a 3:0 observation is 0.25 when there are 10,000 cases and 10,000 controls!)

GWAS interpretation

Often, multiple nearby SNPs show low p-values because they are in linkage disequilibrium and the risk allele is carried by the same individuals.



Hirschhorn and Daly *Nature Reviews Genetics* 2005

Terminology

Diploids: Organisms that possess two homologous copies of each autosomal chromosome

Haplotype: A specific combination of alleles present on a single chromosome

ATCCGA

AGACGC

Genotypes: The complete set of alleles present on both homologous chromosomes

ATCCGA

AGACGC

Linkage disequilibrium (LD)

Note: association tests should work great if the marker you genotyped is actually the disease locus

What if the marker is "nearby" or "correlated" with the disease locus?

Here the concept of LD comes in

The International HapMap Project / 1000 Genomes

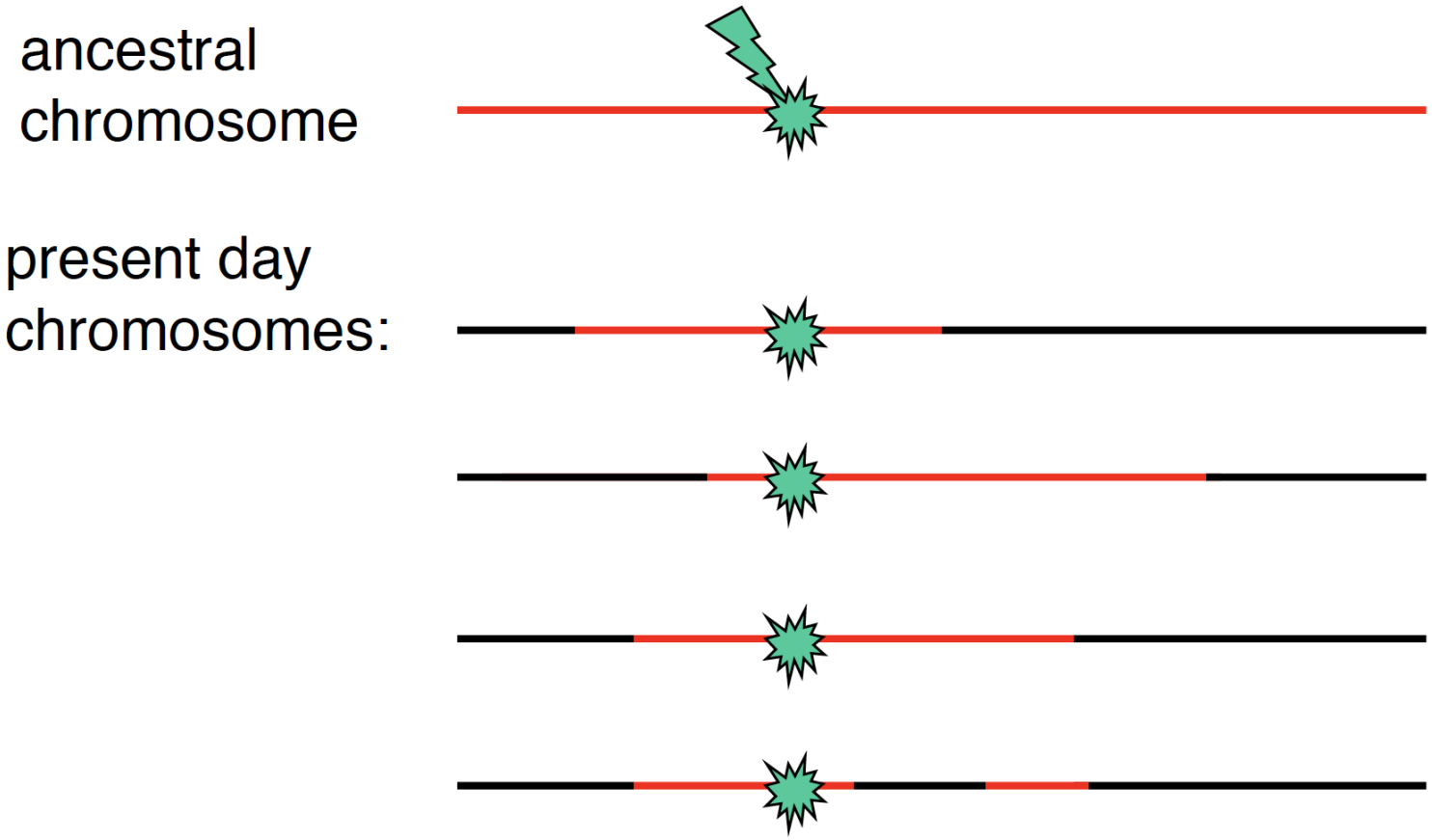
- **Goal:** determine the common patterns of DNA sequence variation (LD among SNPs) in human populations
- Identifies redundancy among SNPs for more efficient disease mapping and pharmacogenetics studies

Human DNA sequence variation

How to measure/describe "patterns" of DNA sequence variation?

How to use these patterns to find disease genes that affect phenotypes?

Human sequence variation



Alleles on the preserved "ancestral background" tend to be in LD

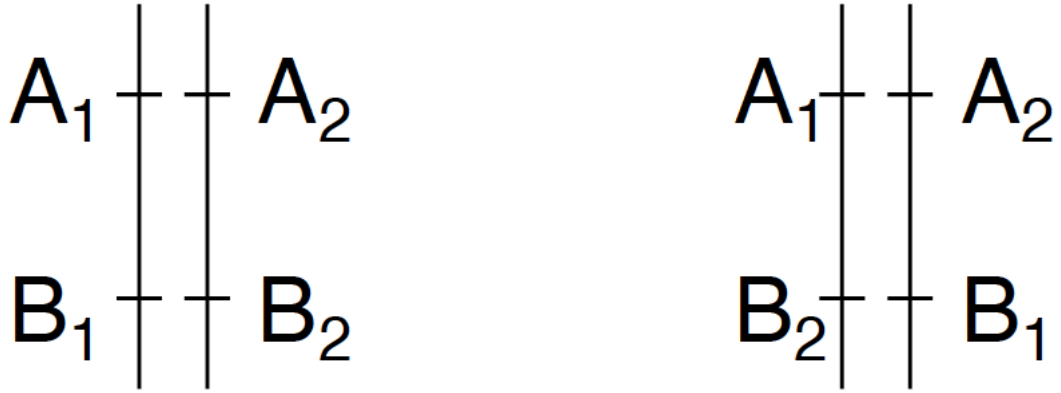
LD involves haplotype frequencies

Focus on pair-wise LD, SNP markers

Genotypes do not necessarily determine haplotypes:

Consider 2-locus genotype A1 A2 B1 B2

Two possible phases :



LD

LD, aka allelic association

For two loci A and B:
LD is said to exist when alleles at A and B tend to co-occur on haplotypes in proportions different than would be expected under statistical independence

How to formally measure LD between alleles at 2 loci?

LD

Example: Consider 2 SNPs:

SNP 1: A 50% C 50%

SNP 2: A 50% G 50%

4 possible haplotypes:

SNP 1	SNP 2	Expected freq
A	A	0.5 * 0.5
A	G	0.5 * 0.5
C	A	0.5 * 0.5
C	G	0.5 * 0.5

But perhaps in your sample you observe only the following:

A A C C A T A T C ... C G A T T ...

and

A A C C C T A T C ... C A A T T ...

e.g.

	A	G	Total
A	0	50	50
C	50	0	50
Total	50	50	100

To measure LD between alleles at 2 biallelic loci

Locus A

A_1, A_2

Locus B

B_1, B_2

	B_1	B_2	
A_1	n_{11}	n_{12}	
A_2	n_{21}	n_{22}	
			2N

Given 2N haplotypes:

Haplotype freq for A_iB_j is

$$h_{ij} = \frac{n_{ij}}{2N}$$

Compare h_{ij} to the frequency expected under no association:

$$p_{A_1}p_{B_1} = \left(\frac{n_{11}+n_{12}}{2N}\right)\left(\frac{n_{11}+n_{21}}{2N}\right)$$

Define the disequilibrium coefficient: $D = h_{11} - p_{A_1}p_{B_1}$

Notes:

1. $D = h_{11} - p_{A1}p_{B1} = h_{22} - p_{A2}p_{B2}$
2. Choice of allele labeling may affect sign but not absolute value of D.

	B_1	B_2	
A_1	n_{11}	n_{12}	
A_2	n_{21}	n_{22}	
			2N

Common LD measures

Disequilibrium coefficient:

$D = h_{11} - p_{A1}\, p_{B1}$

Normalized disequilibrium coefficient:

$D' = D / |D|_{max}$, where

$$|D|_{\max} = \begin{cases} \min(p_{A_1}p_{B_2}, p_{A_2}p_{B_1}) & \text{if } D > 0 \\ \min(p_{A_1}p_{B_1}, p_{A_2}p_{B_2}) & \text{if } D < 0 \end{cases}$$

Range of D' is [-1,1]

Correlation coefficient:

$r^2 = D^2 / (p_{A1}p_{A2}p_{B1}p_{B2})$; a more effective measure of linkage disequilibrium (LD) for assessing allele correlation between markers, indicating the predictive power of an allele at SNP1 for those at SNP2

Describing empirical LD patterns

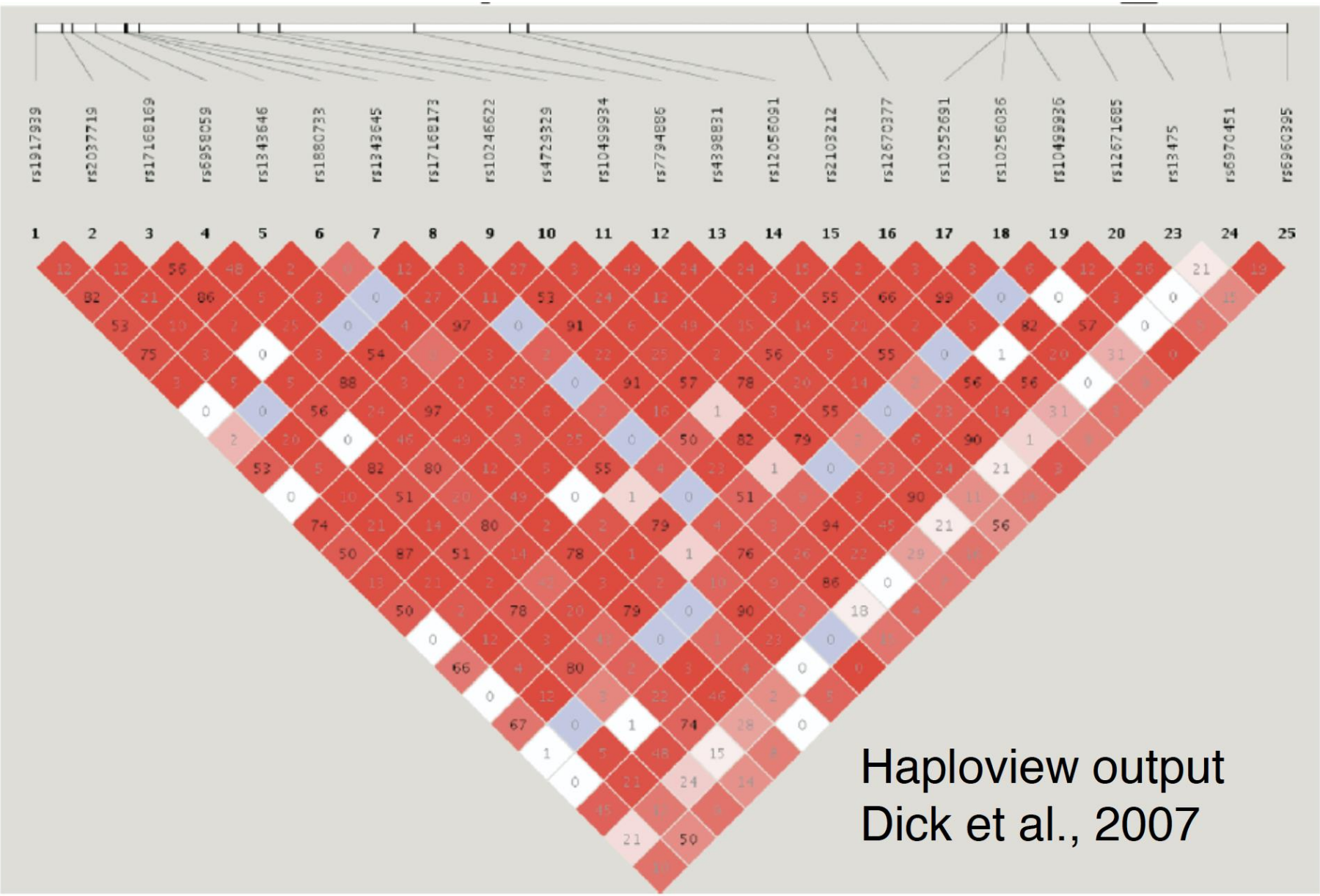


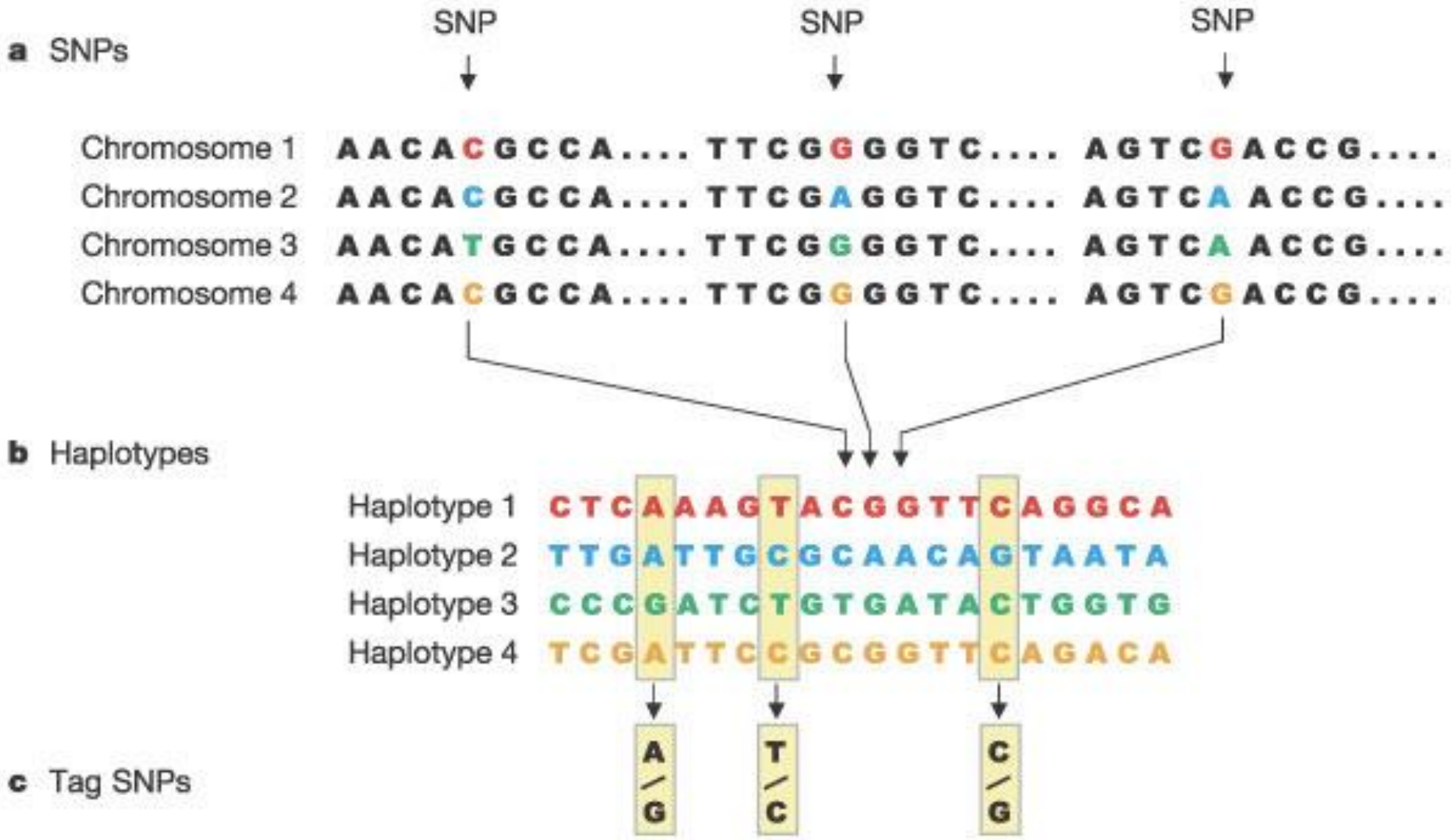
Figure 2. Linkage disequilibrium across the single nucleotide polymorphisms (SNPs) genotyped in and around ACN9. D′ is illustrated by shading, with darker shades indicating higher D′. The r^2 is indicated by the number inside the shaded block.

Using LD in study design

Reference populations - and their LD/haplotype patterns - are used to design “tag SNPs”, impute un-typed variants

- 1000 Genomes Project (1000G)
- Haplotype Reference Consortium (HRC)
- Previously: HapMap: Phase I 2003, Phase II 2007, Phase III

Using LD in study design



The International HapMap Project, Nature 2003

Using LD in study design

A popular LD tag method:

- "r^2 bin tags" (Carlson et al., 2004): greedy algorithm that identifies bins of SNPs such that at least one member of each bin has $r^2 > T$ (threshold) with all bin members
- Note: bin members are not necessarily contiguous

LD patterns inform the design of SNP genotyping arrays, selection of "tag" SNPs

Thus knowledge of LD patterns is important for disease gene mapping

Note: tight linkage between two loci will tend to maintain LD

Decay of LD

After k generations, disequilibrium decays according to

$$D_k = (1-\theta)^k D_0$$

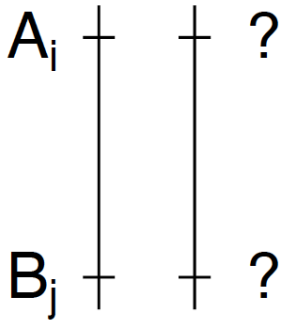
where θ = the recombination fraction (assuming random mating)

$h_{11(1)} = (1-\theta)\, h_{11(0)} + \theta\, p_{A1}p_{B1}$,

so at generation 1,
$D = h_{11} - p_{A1}p_{B1} = (1-\theta)\,(h_{11(0)} - p_{A1}p_{B1})$

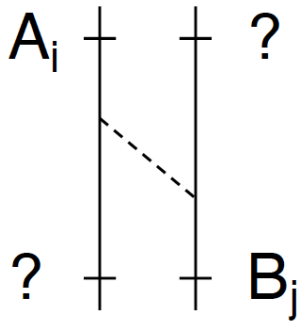
claim: $h_{ij(1)} = (1-\theta)\, h_{ij(0)} + \theta\, p_{Ai} p_{Bj}$,

nonrecombinant



$(1-\theta)\, h_{ij(0)}$

recombinant



$\theta\, p_{Ai} p_{Bj}$

so at generation 1,

$D = h_{11} - p_{A1} p_{B1} = (1-\theta)\,(h_{11(0)} - p_1 q_1) = (1-\theta)\, D_0$

after k generations, get: $D_k = (1-\theta)^k\, D_0$

Decay of LD

Therefore, after k generations

$$h_{ijk} - p_i q_j = (1 - \theta)^k (h_{ijo} - p_i q_j)$$

That is, $D_k = (1 - \theta)^k D_0$

Disequilibrium decays by a factor of $(1 - \theta)$

Half-life of LD

How many generations (t) to reduce by ½?

$$(1 - \theta)^t (h_{ijo} - p_i q_j) = \tfrac{1}{2} (h_{ijo} - p_i q_j)$$

$$(1 - \theta)^t = \tfrac{1}{2}$$

$$t \log (1 - \theta) = \log(0.5)$$

$$t = \log(0.5) / \log(1 - \theta)$$

e.g. if $\theta = 0.5$ (loci are unlinked) then
$t = \log(0.5) / \log(0.5) = 1$

Decay of LD

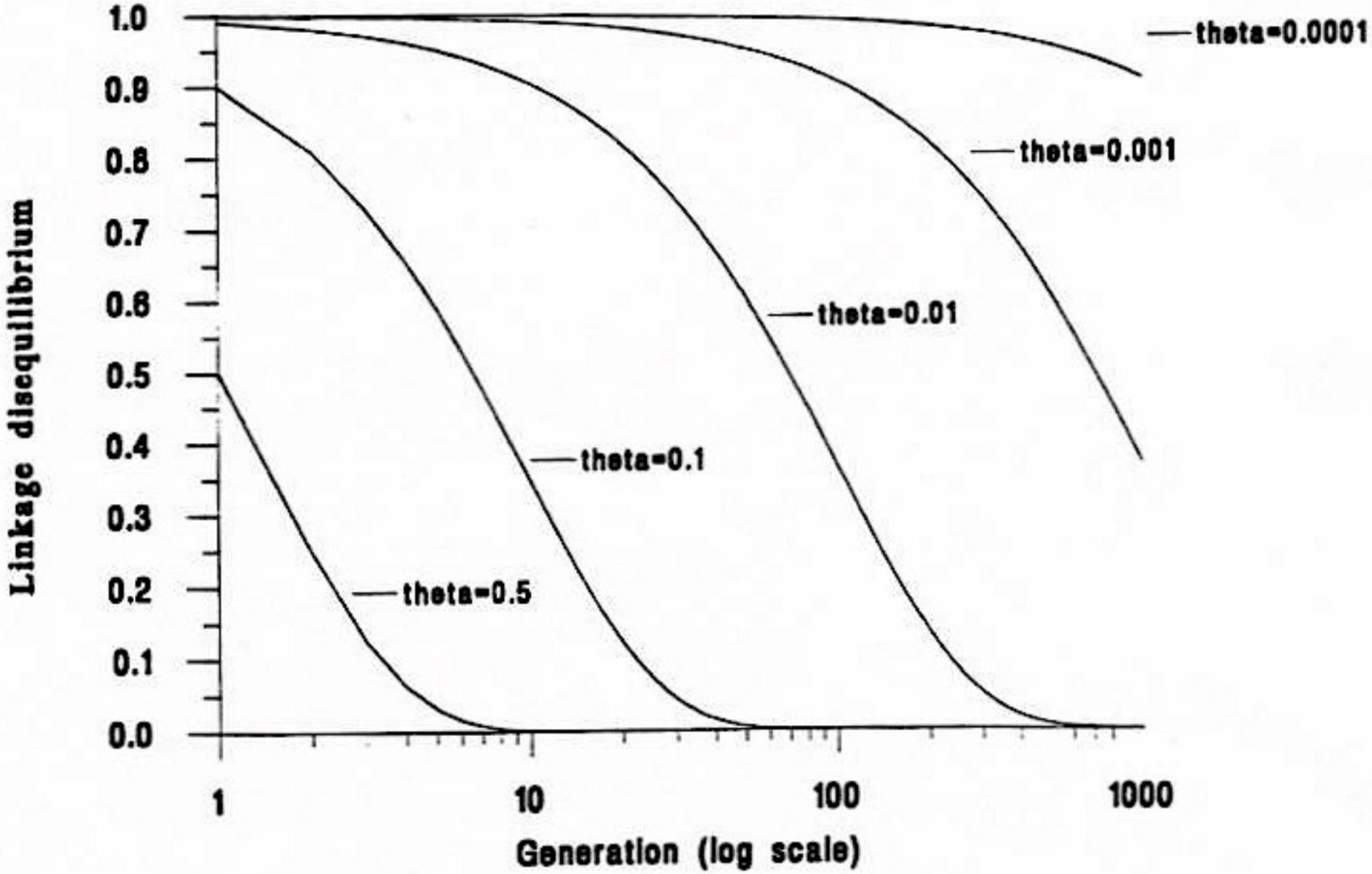
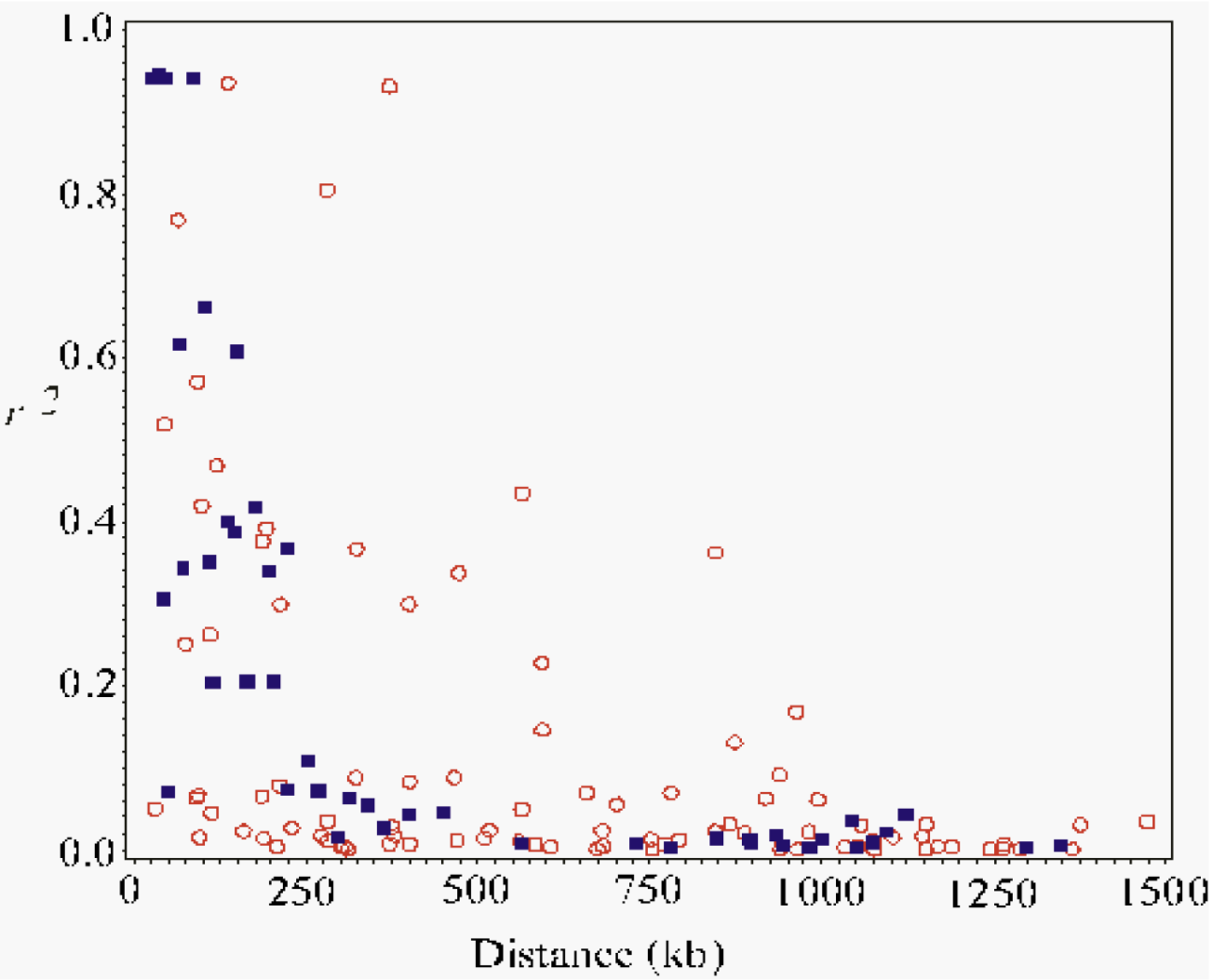


Figure 4.1 Decay of linkage disequilibrium by generation.

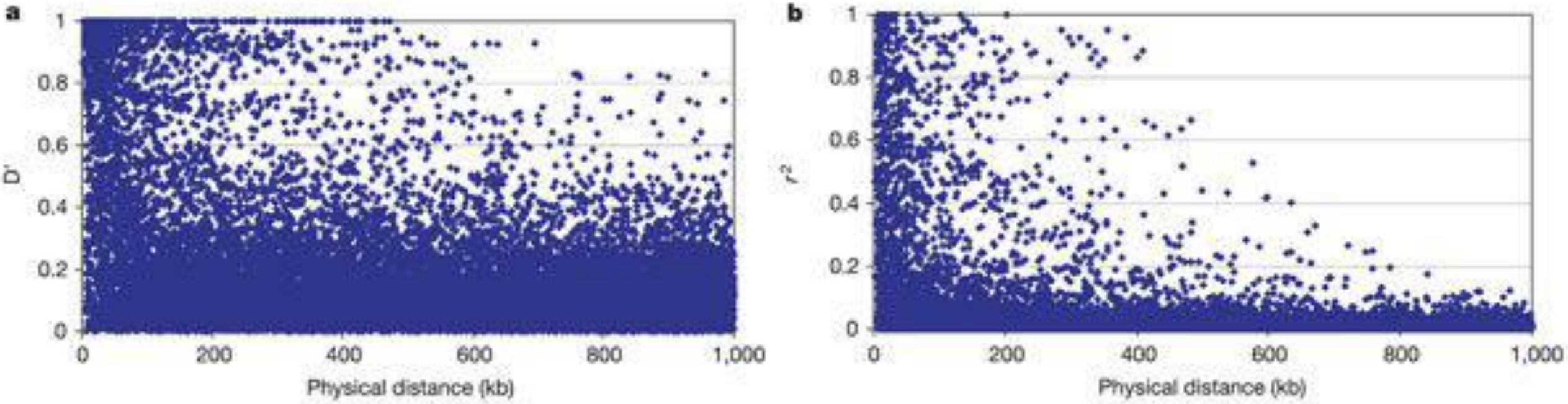
c.f. Pak Sham, Statistics in Human Genetics, Chapter 4

LD is not a simple monotonic function of physical distance



From Taillon-Miller et al., Nat Genet 2000 (O=Xq25, ■=Xq28)

LD is not necessarily a monotonic function of distance
Dawson et al., *Nature* 2002 (chromosome 22)



Where does LD come from?

Potential sources of LD:

1. Linkage between loci
2. Random drift
3. Founder effect
4. Mutation
5. Selection
6. Population admixture / stratification

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Genetic drift generates LD (|D| > 0)

- Via random changes in gamete frequencies
- Smaller isolates: slower decay of LD

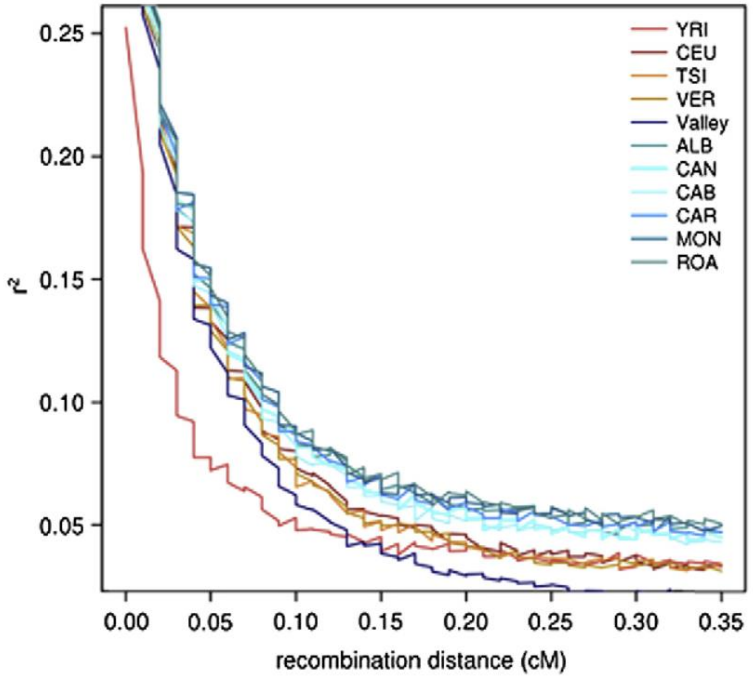


FIGURE 4.3

The decay of linkage disequilibrium as measured by r^2 (Eq. 1.20) with recombination distance in several human populations. The populations labeled ALB, CAN, CAB, CAR, MON, and ROA represent small isolated villages from an Apennine valley in Northwestern Italy. Valley is the conglomerate of all of these isolates, VER represents the Italian population from the Veneto region, TSI the Italian population from Tuscany, CEU the European population, and YRI a large African population.

Modified from Colonna, V., Pistis, G., Bomba, L., Mona, S., Matullo, G., Boano, R., et al., 2013. Small effective population size and genetic homogeneity in the Val Borbera isolate. European Journal of Human Genetics 21, 89–94.

A. Templeton, Human Population Genetics and Genomics, Chapter 4

Where does LD come from?

Potential sources of LD:

1. Linkage between loci
2. Random drift
3. Founder effect
4. Mutation
5. Selection
6. Population admixture / stratification

Suppose have loci A, B, C in that order.

Due to founder effect, suppose sample only 4 haplotypes out of the 8 possible:

A B C

1 1 1

1 2 1

2 1 2

2 2 2

Note:

A and B are in equilibrium

B and C are in equilibrium

A and C are in complete disequilibrium

Disequilibrium not necessarily related to distance!

Where does LD come from?

Potential sources of LD:

1. Linkage between loci
2. Random drift
3. Founder effect
4. Mutation
5. Selection
6. Population admixture / stratification

At the appearance of the mutation, that allele occurs only on one haplotype background.

Where does LD come from?

Potential sources of LD:

1. Linkage between loci
2. Random drift
3. Founder effect
4. Mutation
5. Selection
6. Population admixture / stratification

An example of spurious association due to admixture/stratification:

population 1

9	1	10
81	9	90
90	10	100

chi-square = 0

population 2

25	25	50
25	25	50
50	50	100

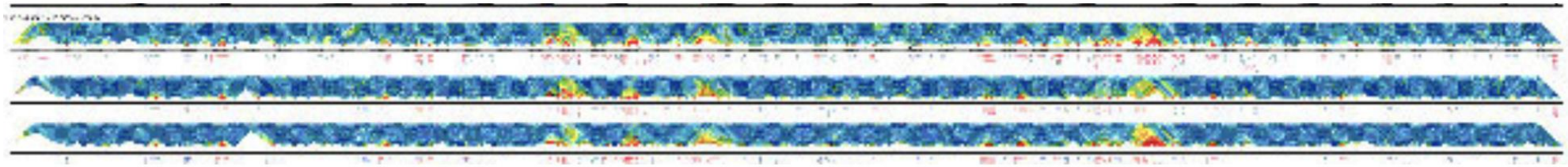
chi-square = 0

combined

34	26	60
106	34	140
140	60	200

chi-square = 7.26
p-value = 0.007

A first generation linkage disequilibrium map of chromosome 22
Dawson et al., Nature 2002



1504 SNPs analyzed in 2 distinct samples

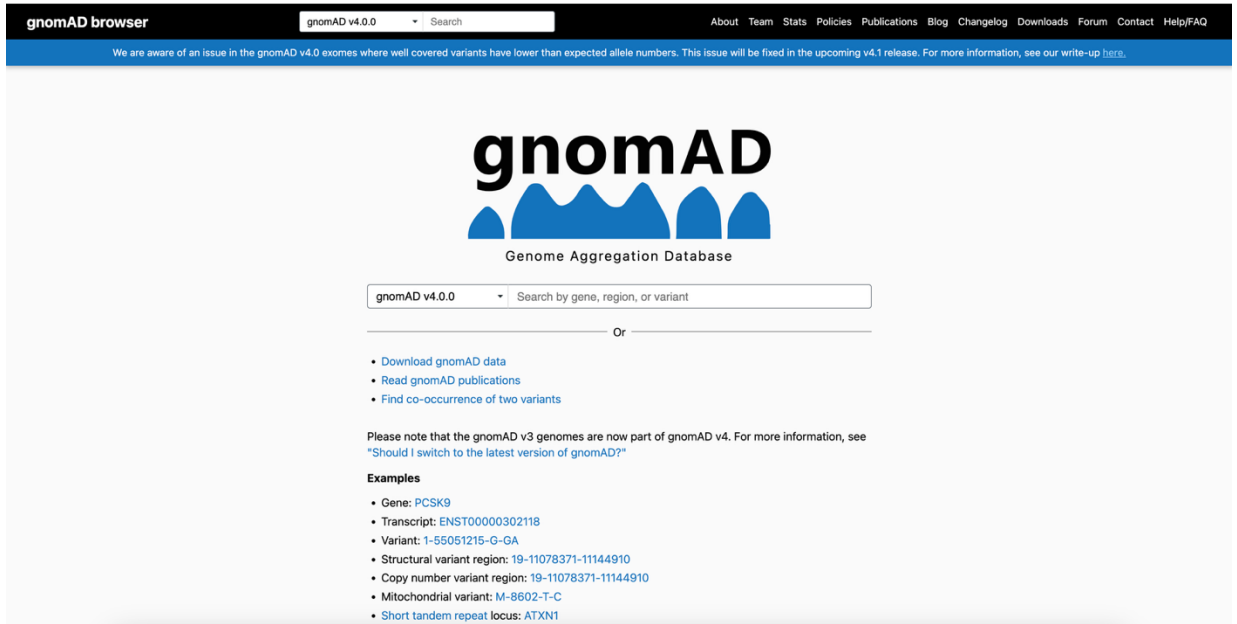
Nature 2003: HapMap I (genome-wide)

Nature 2007

A second generation human haplotype map of over 3.1 million SNPs

The International HapMap Consortium*

We describe the Phase II HapMap, which characterizes over 3.1 million human single nucleotide polymorphisms (SNPs) genotyped in 270 individuals from four geographically diverse populations and includes 25–35% of common SNP variation in the populations surveyed. The map is estimated to capture untyped common variation with an average maximum r^2 of between 0.9 and 0.96 depending on population. We demonstrate that the current generation of commercial genome-wide genotyping products captures common Phase II SNPs with an average maximum r^2 of up to 0.8 in African and up to 0.95 in


gnomAD browser
gnomAD v4.0.0
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We are aware of an issue in the gnomAD v4.0 exomes where well covered variants have lower than expected allele numbers. This issue will be fixed in the upcoming v4.1 release. For more information, see our write-up here.
gnomAD
Genome Aggregation Database
gnomAD v4.0.0
Search by gene, region, or variant
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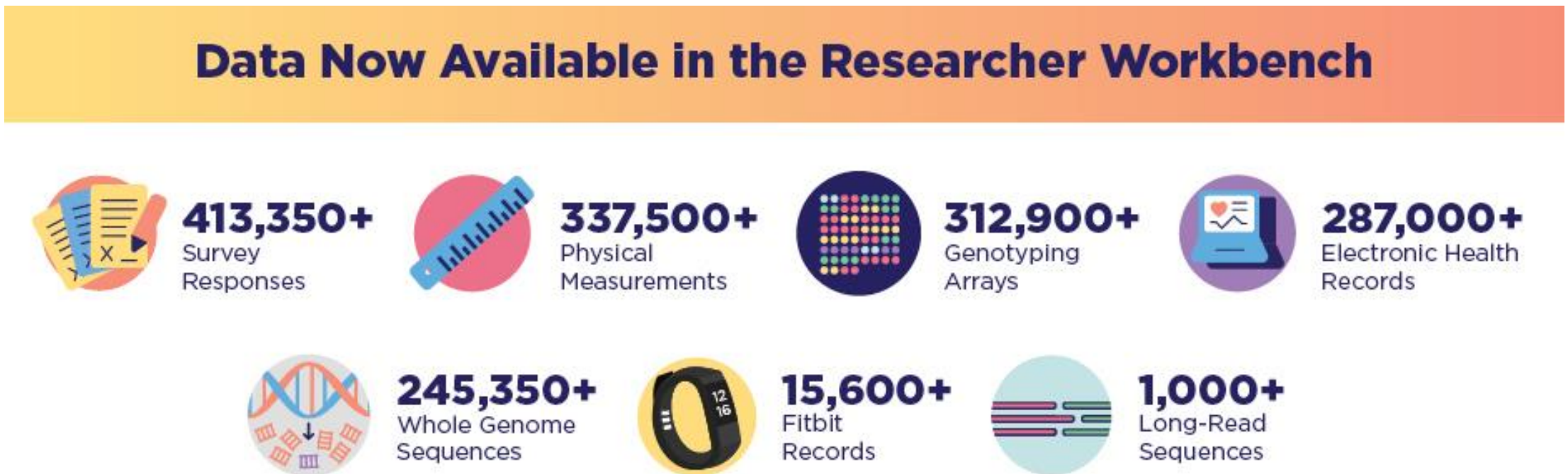
Find co-occurrence of two variants
Please note that the gnomAD v3 genomes are now part of gnomAD v4. For more information, see "Should I switch to the latest version of gnomAD?"
Examples
Gene: PCSK9
Transcript: ENST00000302118
Variant: 1-55051215-G-GA
Structural variant region: 19-11078371-11144910
Copy number variant region: 19-11078371-11144910
Mitochondrial variant: M-8602-T-C
Short tandem repeat locus: ATXN1

biobank uk


Data Now Available in the Researcher Workbench
413,350+
Survey Responses
337,500+
Physical Measurements
312,900+
Genotyping Arrays
287,000+
Electronic Health Records
245,350+
Whole Genome Sequences
15,600+
Fitbit Records
1,000+
Long-Read Sequences